

Hard carbon anodes derived from waste sugarcane bagasse: Temperature regulation of the microstructure and surface oxygen functional groups for high-capacity sodium-ion storage

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Highlights

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Hard carbon anode material is synthesized using sugarcane bagasse.

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Bagasse requires no complex pretreatment, unlike other biomass-derived hard carbons.

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Thermal modulation tuned the porosity and carbonyl groups for Na⁺ storage.

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SBHC-1400 retained 86.5% of its initial capacity after 500 cycles at 200 mA g⁻¹.

Abstract

Although lithium-ion batteries (LIBs) presently constitute the cornerstone of energy storage commerce, they face limitations due to the scarcity and rising expenditure of lithium. As a sustainable substitute, sodium-ion batteries (NIBs) have encouraged consideration because of the profusion and affordability of sodium. However, conventional graphite anodes used in LIBs are unsuitable for NIBs due to poor Na⁺ intercalation. Hard carbons (HCs), particularly those derived from biomass, are auspicious anode materials for NIBs. Some notable features include their structure, expanded interlayer spacing, and high sodium storage capability. In this work, we synthesized a hard carbon anode using sugarcane bagasse, an agricultural waste rich in cellulose, hemicellulose, and lignin. By tuning the carbonization temperature, we optimized the microstructure to enhance Na⁺ storage. The synthesized hard carbon (SBHC-1400) revealed a stable carbon

framework with rich closed micropores and optimum O-containing functional groups, facilitating Na⁺ intercalation and deintercalation. SBHC-1400 demonstrated an excellent reversible capacity of 438.54 mAh g⁻¹ with an initial Coulombic efficiency of 70.4% at 27 mA g⁻¹. Additionally, after 500 cycles at 200 mA g⁻¹, it demonstrated a capacity retention of 86.5%. This work emphasizes the viability of repurposing agricultural waste into value-added energy storage materials, advancing sustainable development.

Introduction

Lithium-ion batteries (LIBs) presently account for the majority share of energy storage landscape, predominantly in consumer electronics and electric cars, as a result of their extended cycle life and high power and energy densities [1,2]. Nevertheless, the escalating energy demand is straining Earth's finite lithium reserves. In contrast, sodium (Na) is abundant and cost-effective, with physical and chemical properties comparable to lithium (Li). This prodigiousness of sodium firmly establishes sodium-ion batteries (NIBs) as promising and sustainable alternatives to LIBs for electrochemical energy storage systems (ESS) [3,4]. Regardless, rapid development of NIBs has been constrained by the lack of suitable anode materials.

In commercial LIBs, graphite serves as the anode material; however, despite the physiochemical similarities between Na and Li, the intercalation of Na⁺ into graphite forms binary graphite intercalation compounds (b-GICs) with unfavorable thermodynamics, a significant hurdle to the widespread adoption of NIBs [5]. Carbon is one of the most copious elements on Earth, and because of its fascinating properties, it has been widely used in many fields [[6], [7], [8], [9]]. Many carbon materials exist, including fullerenes, graphene, carbon nanotubes, *etc.* Among the numerous carbon materials, hard carbon, which has a stable carbon skeleton and enlarged interlayer spacing, is one of the most economically viable and best anode materials for NIBs [10]. Unlike graphite, hard carbons (HCs) possess copious short-range graphite microcrystals with a long-range amorphous configuration that facilitates electron transport through the graphitic channel and delivers ample accessible actives for Na⁺ storage [11,12]. The Na storage process is typically described by two regions in the charge-discharge profile. That is, during sodiation, Na⁺ inserts into the carbon layers and/or adsorbed on the defect sites in the sloped region (0.1–1 V (vs. Na⁺/Na)), while in the plateau region (<0.1 V), the micropores are filled with Na⁺ [13].

Biomass-derived hard carbons have garnered substantial interest due to their sustainability and cost-effectiveness, as opposed to coal-derived hard carbons which require energy intensive processing, involve hazardous chemical treatments, and are inherently unsustainable [[14], [15], [16]]. In the course of the high-temperature

carbonization of the biomass precursor, the carbon layer forms defective graphene nanosheets turbostratically stacked together, comprising surface defects, graphitic domains, and nanopores [17,18]. Low Initial Coulombic Efficiency (ICE) arises from the irreversible entrapment of Na^+ at defect sites. Introducing heteroatom dopants into the carbon structure (e.g., N, O, P, S) can augment the Na^+ transfer kinetics, provide supplementary active sites, and increase the interlayer distance [19]. At higher carbonization temperatures, more closed micropores are formed, maximizing plateau capacity for sloping capacity [20]. Thus, the electrochemical characteristics for reversible Na^+ storage are affected by the carbon precursor, carbonization temperature, morphology, and heteroatom doping [[21], [22], [23]].

Sugarcane is one of the most prodigious biomass types, responsible for about 80% of global sugar production [24]. The fibrous substance left over after sugar juice is extracted is called bagasse (comprising cellulose, hemicellulose, lignin, moisture, and ash), and every ton of sugarcane yields about 280 kg of bagasse [25,26]. Bagasse is a waste product with low commercial worth commonly incinerated in the open air, resulting in greenhouse gas emissions and pollution. A key tenet of sustainable development is recycling/repurposing waste products into value-added products, and bagasse is a prospective sustainable and economical precursor for hard carbon, making it advantageous over other biomass types. Another advantage of bagasse over other biomass types such as bamboo is that it does not require complex pre-oxidation treatments or multi-step pyrolysis protocols commonly employed to introduce oxygen-containing functional groups in bamboo-derived hard carbons. Bagasse exhibits an optimum equilibrium between cellulose and hemicellulose/lignin, making it suitable as a precursor for fabricating hard carbon electrode materials [27,28].

Herein, we report fabricating a hard carbon anode material with outstanding electrochemical characteristics using bagasse as the precursor. The temperature can modulate the microstructure of the synthesized hard carbon. Moreover, by adjusting the temperature, the closed micropores, which enhance Na^+ storage in the plateau region, and

carbonyl groups ($\text{C}=\text{O}$), which function as readily available active sites for reversible Na^+ adsorption to increase sloping capacity, can be regulated. The synthesized anode material SBHC-1400 showed exceptional electrochemical performances with high reversible specific capacity ($438.54 \text{ mAh g}^{-1}$) at 27 mA g^{-1} and ICE (70.4%). After 500 cycles at 200 mA g^{-1} , the 86.5% of the initial capacity was preserved.

Hard carbon synthesis

The graphic representation of the synthesis is illustrated in Fig. 1. Sugarcane bagasse is first soaked in 2 M H₂SO₄ overnight to effectively remove inorganic impurities. After acid treatment, it is thoroughly washed with ultrapure water and dried at 80 °C. In addition to deashing, the sulfuric acid partially hydrolyzes hemicellulose and cellulose, facilitating the release of volatile compounds and promoting pore formation during subsequent carbonization. Moreover, acid pretreatment suppresses

Physical characterization

Fig. 2 reveals the morphology of the hard carbons using Scanning Electron Microscopy (SEM) and High-Resolution Transmission Electron Microscopy (HRTEM). From Fig. 2(a–h), the SEM images show that all the synthesized hard carbons had a bulky and nanosheet-like morphology, showing that regardless of carbonization temperature, the morphology barely changed. The nanosheet-like morphology is beneficial because it provides copious active sites for electrochemical reactions and reduces Na⁺ diffusion

Conclusion

In conclusion, hard carbon materials derived from sugarcane bagasse were fabricated and evaluated as anode materials for NIBs. The carbonization temperature markedly impacted the microstructural evolution and electrochemical behavior of the hard carbon materials, modulating parameters such as porosity, defect density and Na⁺ storage behavior. With elevated temperatures, the specific surface area, pore volume, structural defects, and oxygen-containing functional groups progressively decreased,

CRedit authorship contribution statement

Daniel Kobina Sam: Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Conceptualization. **Jiaze Lv:** Writing – review & editing, Validation, Software, Methodology, Investigation. **Aaron S. Pittman:** Writing – review & editing, Validation, Methodology, Investigation. **Yan Cao:** Writing – review & editing, Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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